SECURITY ANALYSIS (BBAT35F97) · TAPMI · BBA(H) SEMESTER V

AI-AUGMENTED SECURITY ANALYSIS

Valuing Hindustan Unilever, Sun Pharma and ICICI Bank, and building a long-term portfolio from them

FMCG
31% OF P

Hindustan Unilever

β 0.60 · CAGR 9.6%

PHARMA
19% OF P

Sun Pharma

β 0.65 · CAGR 8.7%

PRIVATE BANK
50% OF P

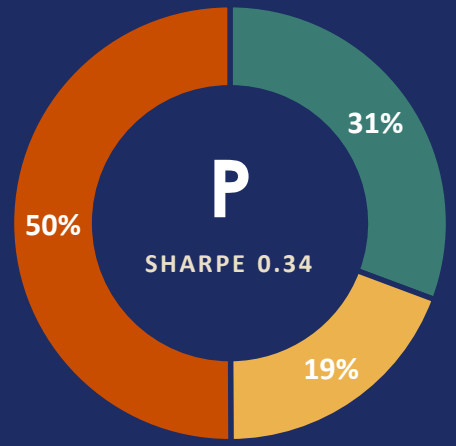
ICICI Bank

β 1.29 · CAGR 19.1%

GROUP PROJECT · DATA AS OF 13 OCT 2026

HUL · SUN PHARMA · ICICI BANK

THE PORTFOLIO WE BUILT



₹64.8 lakh
MEDIAN 15-YEAR CORPUS FROM A ₹10,000 MONTHLY SIP

44 / 47
AI-EXTRACTED FIGURES WITHIN 1% OF YAHOO AND SCREENER

0 of 8
TRADING RULES THAT BEAT BUY-AND-HOLD AFTER COSTS

THREE SECTORS, THREE DIFFERENT DRIVERS

Top-down: an investment-led economy, then three businesses driven by different demand.

7.8%	+11.9%	5.25%	5.52%
REAL GDP GROWTH, Q1 FY27	INVESTMENT (GFCF) GROWTH	RBI REPO RATE	91-DAY T-BILL RATE

FMCG

Hindustan Unilever

DEFENSIVE ANCHOR

Lowest beta in the weakest sector: out-of-favour quality.

β 0.60 · CAGR 9.6%

PHARMA

Sun Pharma

DEFENSIVE GROWTH

Demand ignores the cycle; US sales earn dollars.

β 0.65 · CAGR 8.7%

PRIVATE BANK

ICICI Bank

GROWTH ENGINE

The direct play on investment-led growth: loans +19.6%.

β 1.29 · CAGR 19.1%

Sources: MOSPI, RBI, Zerodha Aftermarket Report (12 Oct 2026). Full funnel: docs/macro_industry.md.

WHY THEY BELONG TOGETHER

0.16–0.24

PAIRWISE CORRELATIONS, 2016–2026

Betas of 0.60, 0.65, 1.29: two defensives and one cyclical.

WHAT WE REJECTED

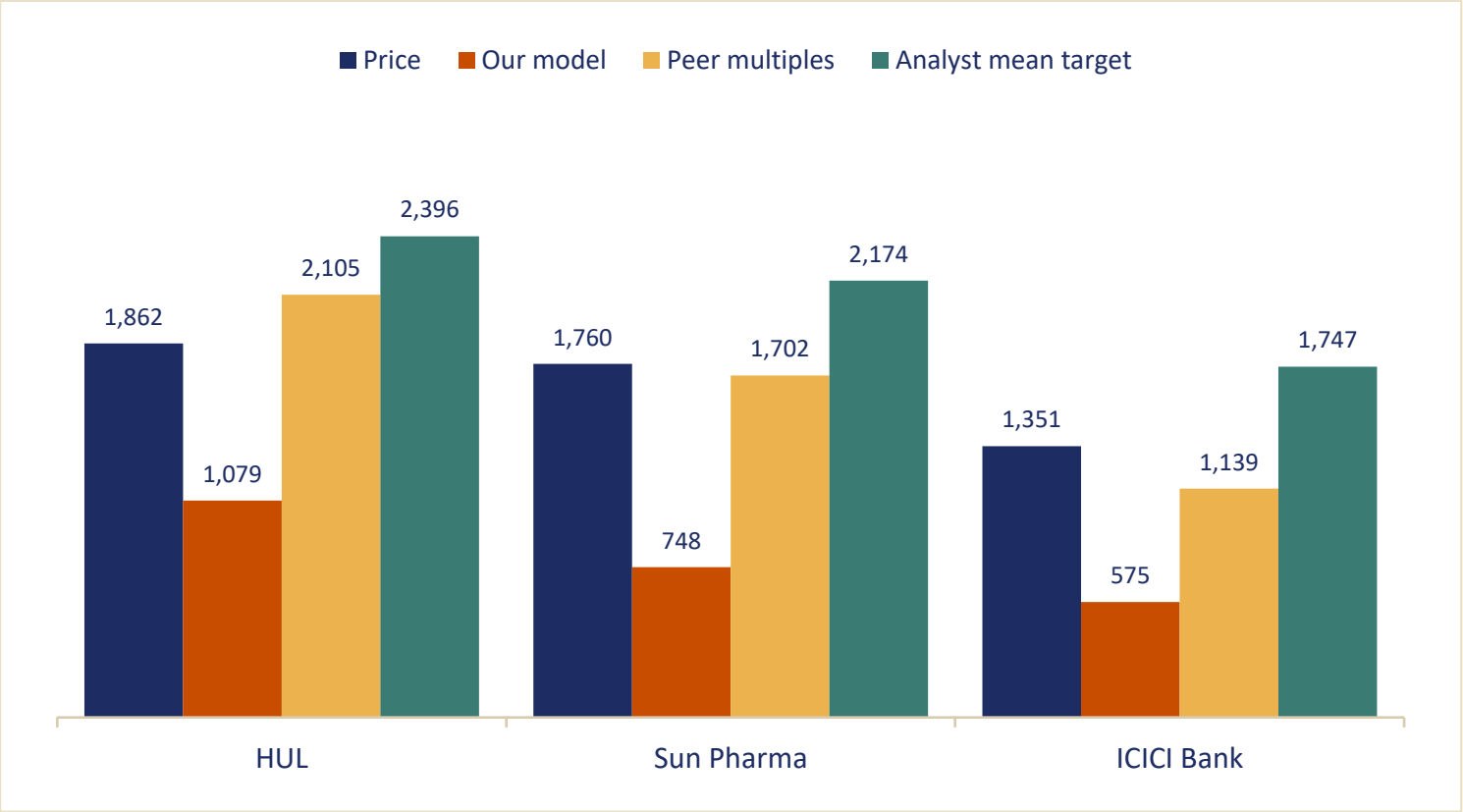
MEDIA
Strongest sector (+11.2% vs 200-day), but its certainty equivalent at A = 4 is −8% to −38%: chasing it is recency bias.

INFOSYS
IT is 8.9% below its 200-day; Sun keeps the dollar hedge at a lower beta.

HDFC BANK
The 2023 merger breaks its history.

PRICED FOR MORE GROWTH THAN WE CAN JUSTIFY

Intrinsic value from our Excel models at CAPM discount rates, against today's price and the analysts' targets.



−14%

BLENDED VS PRICE

Hindustan Unilever

Price implies 7.4% growth a year for ever, at r = 9.8%

−30%

BLENDED VS PRICE

Sun Pharma

Price implies 8.6% growth a year for ever, at r = 10.1%

−37%

BLENDED VS PRICE

ICICI Bank

Price implies 13.9% growth a year for ever, at r = 14.7%

ICICI grew loans 19.6% last quarter. The gap is mostly our 14.7% cost of equity (β 1.29), not unrealistic growth.

Two-stage DDM (HUL), two-stage residual income (Sun Pharma, ICICI Bank), peer P/E and ROE-adjusted P/B: workbook/valuation.xlsx. Blended = average of model and peer values. Analyst targets: Yahoo Finance consensus.

FIVE AI TASKS, EACH CHECKED AGAINST A BENCHMARK

AI reads and labels; Python computes every number. Each task reports how far it can be trusted.

01 RATIO EXTRACTION

WHAT
Claude reads the FY26 annual reports and returns line items as JSON.

CHECKED AGAINST
Yahoo Finance and Screener.

44 / 47

within 1%; 0 extraction errors

02 SENTIMENT

WHAT
FinBERT scores 2,428 headlines and 9,133 call sentences.

CHECKED AGAINST
Management vs analysts on every call.

+0.28

HUL tone gap: management far more upbeat

03 BIAS & FRAMING

WHAT
An LLM tags 120 headlines for framing, authority, herding and hype.

CHECKED AGAINST
70% agreement with FinBERT.

41%

carry a bias cue; 14% aren't about the company

04 FORECASTING

WHAT
Moving-average and RSI rules, an earnings event study, analyst consensus.

CHECKED AGAINST
Buy-and-hold, after a 0.25% trading cost.

0 of 8

trading rules beat buy-and-hold

05 RISK HEATMAPS

WHAT
Correlation, rolling correlation, risk contribution, stress tests.

CHECKED AGAINST
Four real sell-offs since 2020.

57%

of portfolio risk is ICICI, at 50% weight

Prompts: docs/ai_prompts.md. Extraction: data/ai/ratio_extraction.json (page references for every figure). FinBERT: ProsusAI/finbert. Headlines: Google News RSS, last 12 months.

SEE IT LIVE

Two minutes, four stops on our dashboard.

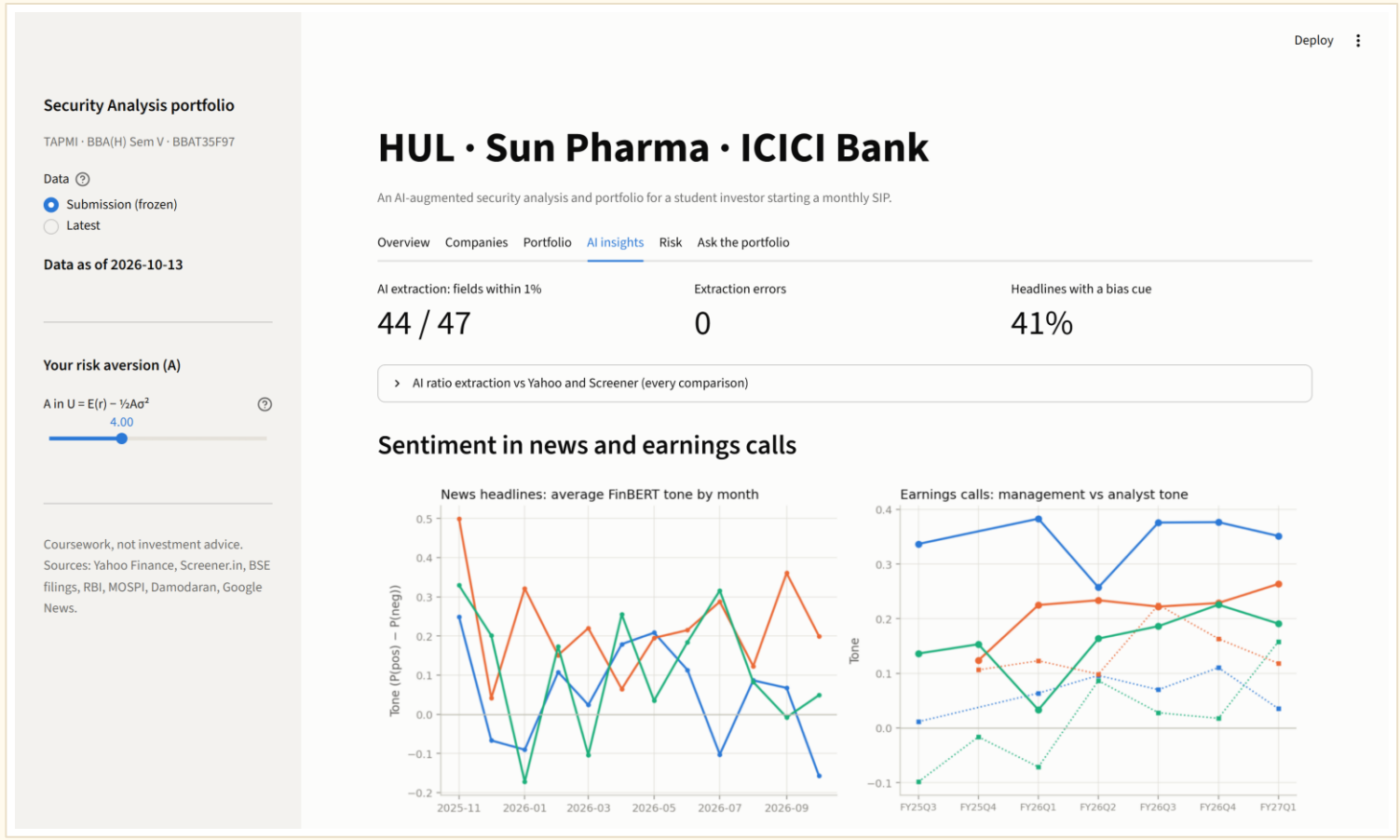
01 **AI INSIGHTS**
Monthly sentiment; news framing

02 **RISK**
Correlation heatmap; stress tests

03 **ASK THE PORTFOLIO**
One question, answered with its sources

04 **YOUR OWN PORTFOLIO**
Slide A from 4 to 2: your share in P goes from 44% to 87%

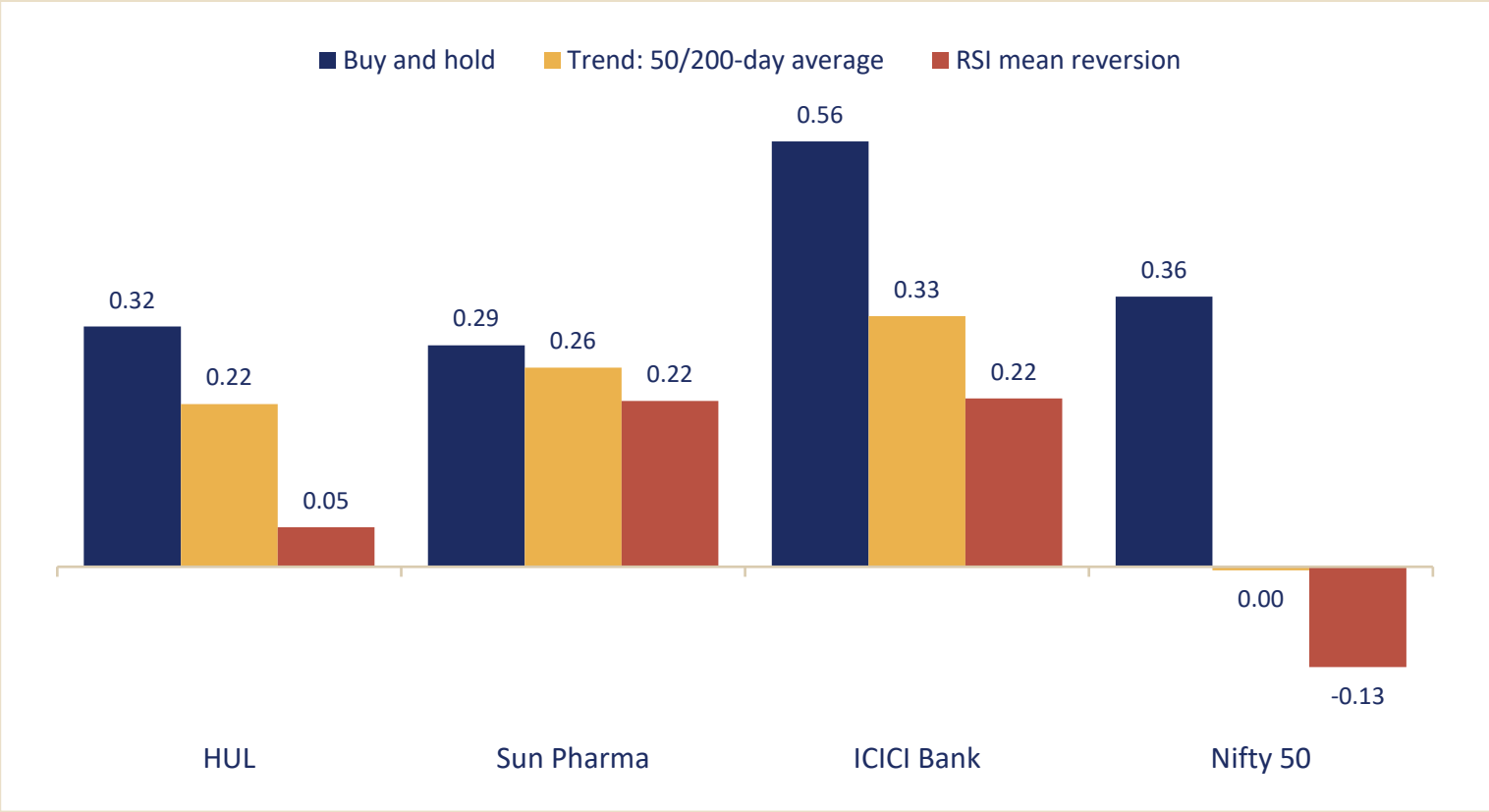
LOCAL COPY
streamlit run app.py



IF ANYTHING TAKES MORE THAN 10 SECONDS
1 Live link → 2 Local copy → 3 Recorded walkthrough → 4 Screenshots (A3)

TECHNICAL ANALYSIS LOSES AFTER COSTS

The brief asked: is technical analysis possible? Sharpe ratio 2016–2026, after a 0.25% round-trip cost.



Prices: Yahoo Finance, 2016–2026. Earnings surprises: Yahoo Finance earnings dates. Figure: data/submission/figures/ta.png.

WEAK-FORM EFFICIENCY

0 OF 8

RULES THAT BEAT BUY-AND-HOLD

The rules cut drawdowns, so they are risk controls, not return engines.

SEMI-STRONG: NEWS IS PRICED FAST

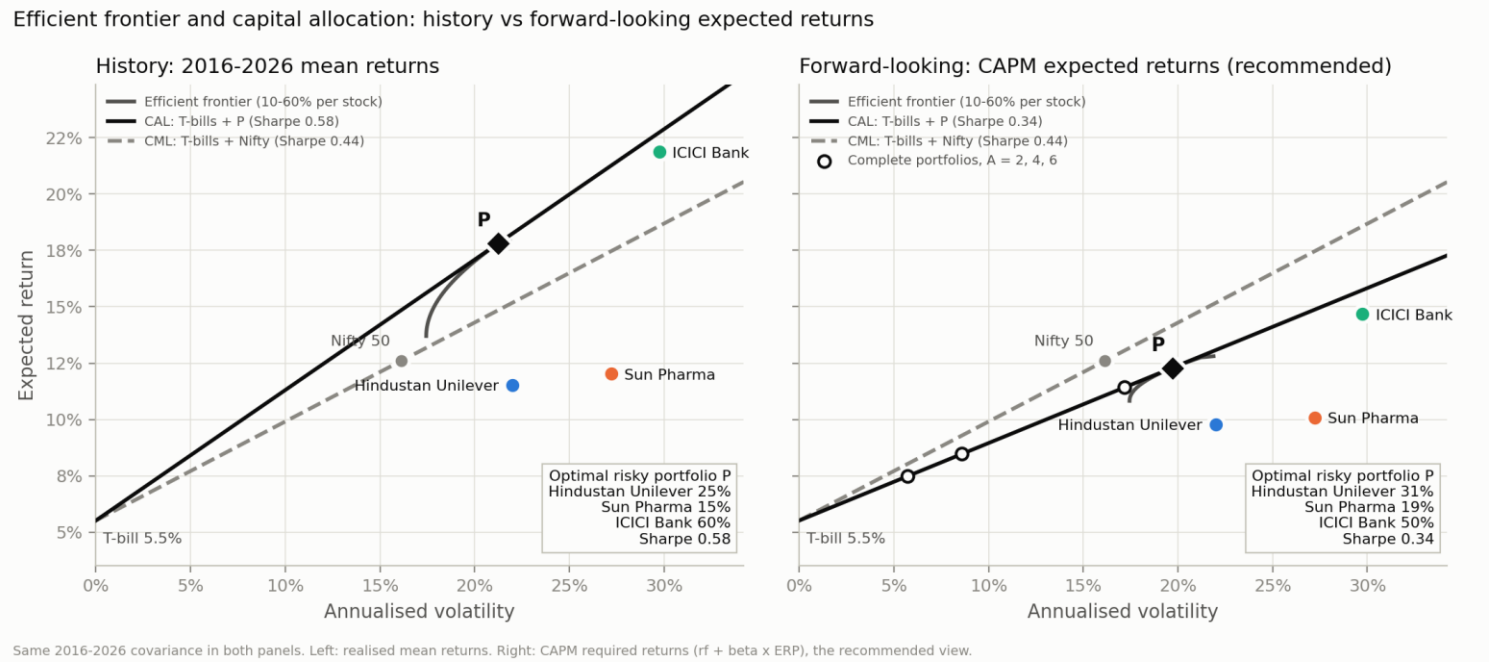
+1.2% / −1.6%

3-DAY ABNORMAL RETURN AFTER EARNINGS BEATS / MISSES

72 results announcements since 2016; market-model CAR over days −1 to +1.

31 / 19 / 50, SIZED TO EACH INVESTOR

Forward-looking (CAPM) expected returns, 10–60% per stock. History alone would put 60% in last decade's winner.



87%
IN P

AGGRESSIVE · A = 2
E(r) 11.4% · σ 17.2%
Certainty equivalent 8.5%

44%
IN P

BALANCED · A = 4
E(r) 8.5% · σ 8.6%
Certainty equivalent 7.0%

29%
IN P

CONSERVATIVE · A = 6
E(r) 7.5% · σ 5.7%
Certainty equivalent 6.5%

SHARPE 0.34 vs 0.44

Three stocks don't beat the index on risk-adjusted terms, so the SIP pairs them with a Nifty core. Out of sample (2023–26): forward P +7.9%/yr, Nifty +7.4%, history-based P +3.2%.

OVER THE YEARS

₹10,000 a month for 15 years, under our Investment Policy Statement.

RETURN
Beat 4% inflation clearly; target ₹75 lakh

RISK
Balanced (A = 4) unless the questionnaire says otherwise

LIQUIDITY
6 months of expenses kept outside the portfolio

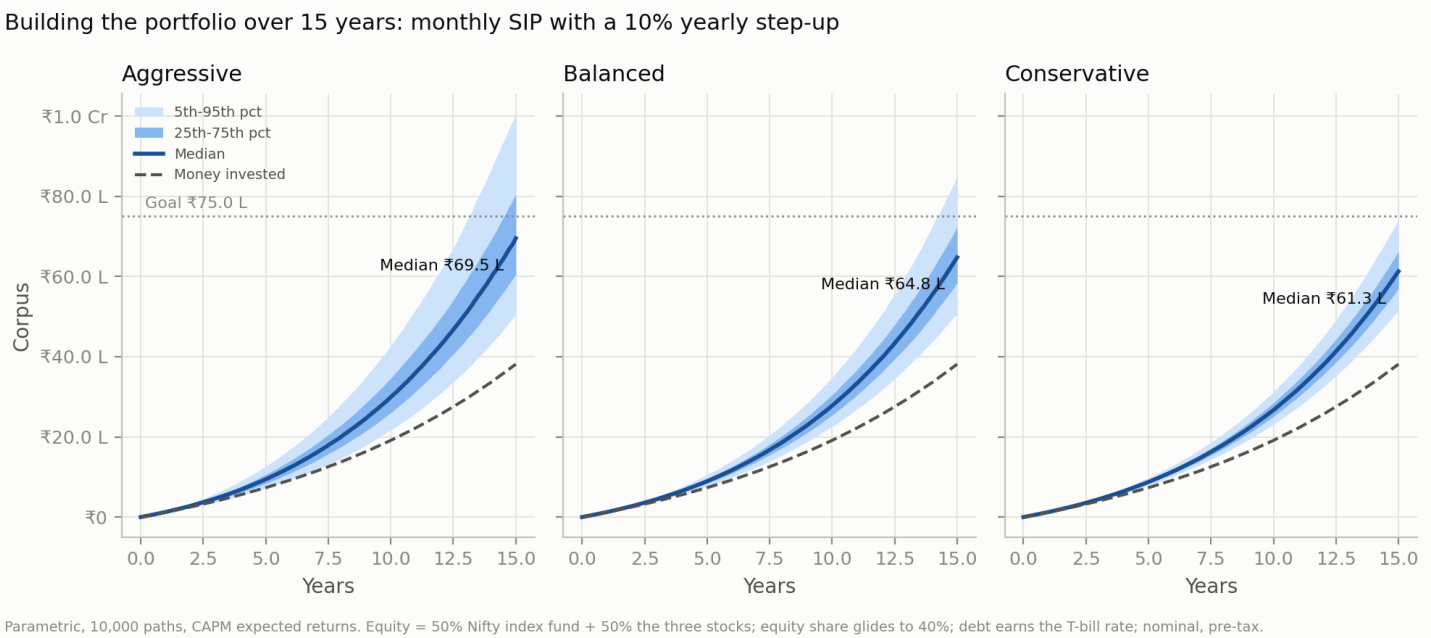
REBALANCE
Once a year, with new SIP money first

BEHAVIOUR
No selling in the month after a 20% fall

₹38.1 lakh
INVESTED (10% STEP-UP A YEAR)

₹64.8 lakh
MEDIAN CORPUS, BALANCED

19% / 11%
CHANCE OF ₹75 LAKH (TWO MODELS)



50% Nifty 50 index fund + 50% our three stocks, gliding to 40% equity. Parametric and bootstrap Monte Carlo, 10,000 paths, pre-tax. Aggressive: 36% / 29% chance of ₹75 lakh.

OUR CALL

Range: our blended value to the analysts' mean target.

HINDUSTAN UNILEVER

BUY

₹1,592–2,396 · price ₹1,862

SUN PHARMA

HOLD

₹1,225–2,174 · price ₹1,760

ICICI BANK

HOLD

₹857–1,747 · price ₹1,351

BUY THE PORTFOLIO SLOWLY, THROUGH THE SIP

Good businesses at full prices: own them, but let monthly buying average the entry price.

RISKS WE ARE WATCHING

01 **CONCENTRATION** ICICI is 50% of the weight but 57% of the risk.

02 **VALUATION** Today's prices assume 7.4% to 13.9% growth a year, for ever.

03 **REGULATION** Proposed caps on trade margins for anti-cancer drugs (Sun Pharma).

LIMITATIONS

One risk-free rate for 2016–26 · Nifty total return estimated (+1.22%/yr dividends) · Sun Pharma chosen after its run · pre-tax, no fees · coursework, not investment advice.

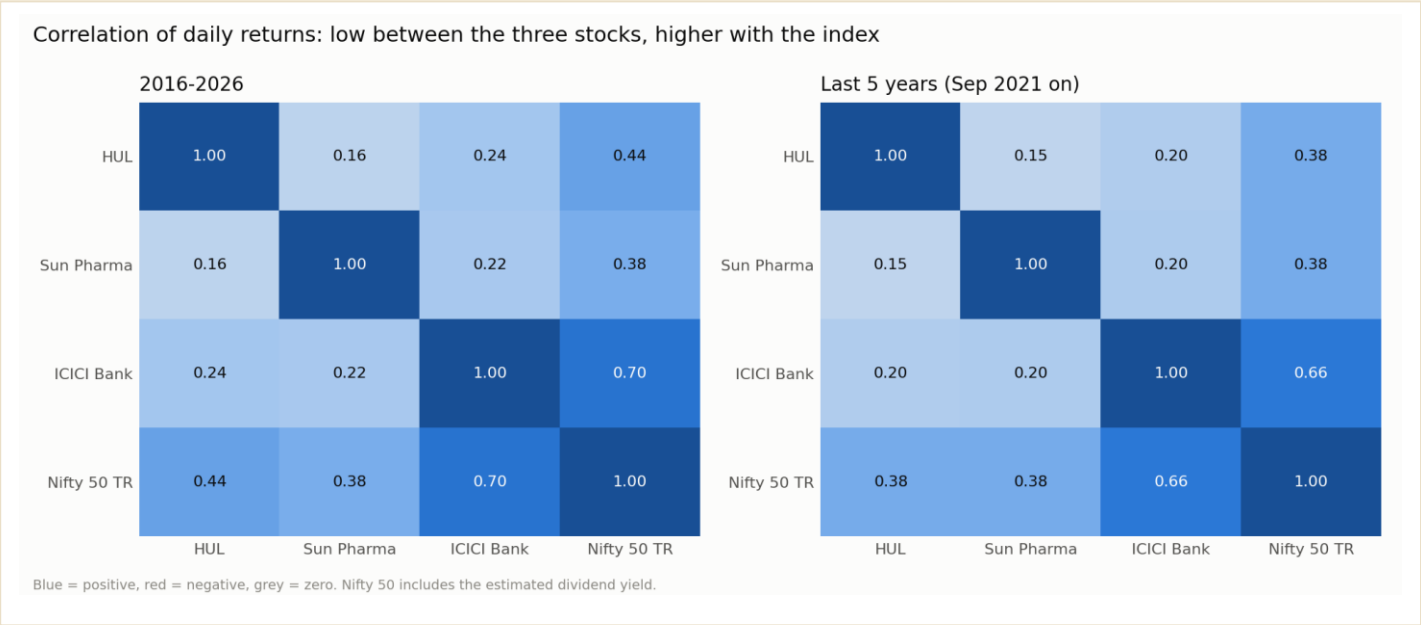
METHODS AND ASSUMPTIONS

Every number in this deck is computed from these choices; the code is in our GitHub repository.

Item	Choice	Source
Prices	Daily adjusted closes from 2016; HUL demerger (Dec 2025) adjusted	Yahoo Finance; NSE special pre-open
Risk-free rate	5.52%, 91-day T-bill	RBI, 1 Oct 2026
Equity risk premium	7.08% (India)	Damodaran, 5 Jan 2026
Benchmark	Nifty 50 price index + 1.22%/yr estimated dividends	Nifty BeES vs index gap, 2016–2026
Valuation	Two-stage DDM (HUL); two-stage residual income (Sun Pharma, ICICI); peer multiples; r × g grids	workbook/valuation.xlsx
Optimisation	Maximum Sharpe ratio, 10–60% per stock, CAPM expected returns (rf + β × ERP)	scipy SLSQP
Complete portfolio	$y^* = (E[rP] - rf) / (A\sigma^2)$, capped at 100%; A = 2, 4, 6	BKM Ch 6
Monte Carlo	10,000 paths, 1 year; normal (Cholesky), 10-day block bootstrap, EWMA λ = 0.94	numpy, seed 42
SIP	₹10,000/month, +10%/yr, 15 years; 50% Nifty core; glide to 40% equity	Pre-tax, no fees
AI	FinBERT sentiment; Claude for extraction, aspects and headline tagging	docs/ai_prompts.md

WHERE THE RISK SITS

Correlations of daily returns, and what happened to the portfolio in four real sell-offs.



Episode	Our P	Nifty 50
COVID crash (Feb-Mar 2020)	**−34.6%**	−37.2%
Rate-hike sell-off (Jan-Jun 2022)	**−12.3%**	−16.1%
Election-result shock (4 Jun 2024)	**−2.3%**	−5.9%
2026 drawdown (02 Jan 2026 to 13 Oct 2026)	**−5.1%**	−13.7%

Our portfolio fell less than the Nifty in 4 of 4 episodes.

32%

HUL
SHARE OF PORTFOLIO RISK

11%

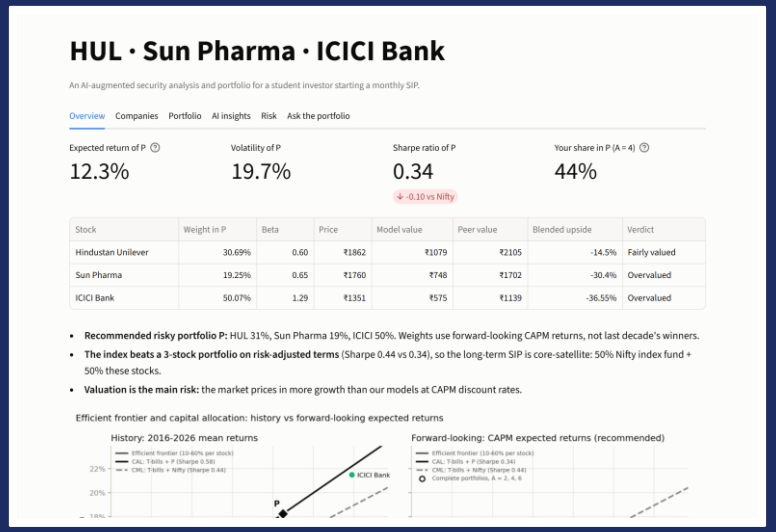
SUN PHARMA
SHARE OF PORTFOLIO RISK

57%

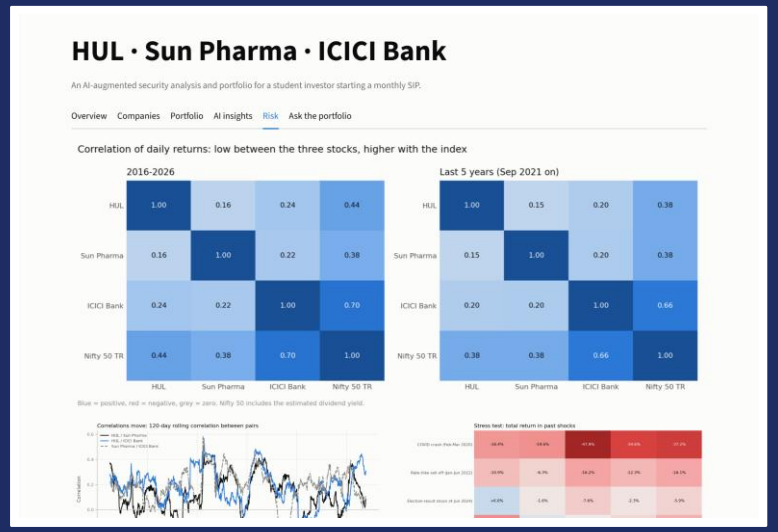
ICICI BANK
SHARE OF PORTFOLIO RISK

THE DASHBOARD, IF THE DEMO FAILS

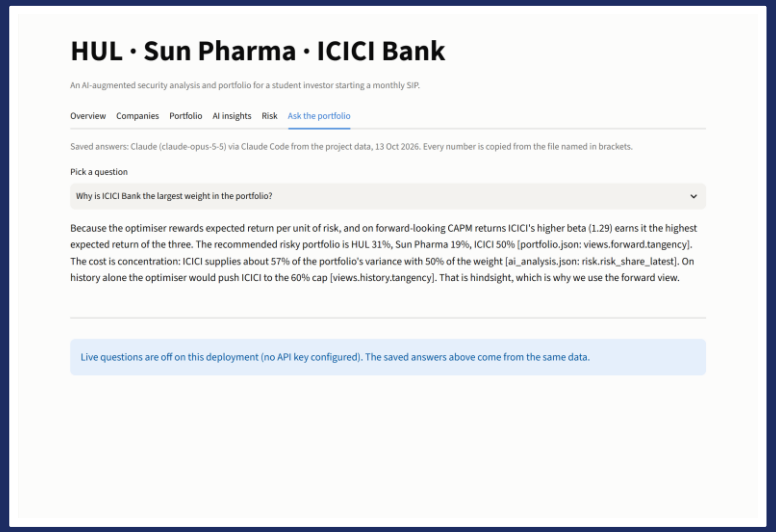
Rung 4 of the backup ladder: the same screens as the live demo.



OVERVIEW: P, VALUATION, VERDICTS



RISK: HEATMAPS AND STRESS



ASK THE PORTFOLIO: SAVED ANSWERS

DATA SOURCES AND AI PROMPTS

Free and public sources only; every AI prompt is saved with its output.

DATA

- Prices, statements, analyst consensus: Yahoo Finance (yfinance)
- Ten-year financials: Screener.in public pages
- Annual reports and earnings-call transcripts: BSE filings and company IR sites
- Risk-free rate and repo rate: Reserve Bank of India
- GDP and demand mix: MOSPI press note, 31 Aug 2026
- Equity risk premium: Damodaran, India, Jan 2026
- Headlines: Google News RSS, 12 months
- Sector momentum: Zerodha Aftermarket Report, 12 Oct 2026

AI AND METHOD

- FinBERT: ProsusAI/finbert (Hugging Face)
- Claude (Anthropic): ratio extraction, concall aspects, headline tagging
- Prompts P1–P5 and outputs: docs/ai_prompts.md, data/ai/
- Bodie, Kane & Marcus, Investments: Ch 6, 7, 12, 17, 18, 24
- Valuation workbook: workbook/valuation.xlsx (live formulas)
- Code, tests and the nightly refresh: GitHub repository